

Sera Nolin

Rendering Researcher

Research-oriented graphics programmer investigating visibility, material representation, GPU scheduling, and practical renderer architecture.

loc: Lakeview Research Campus
env: Linux
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git: github.com/fake-sera-nolin

/// SKILLS

Build:

CMake Ninja Git Linux

Graphics:

Vulkan OpenGL GLSL SPIR-V PBR
Shadow Mapping

Mathematics:

Linear Algebra Geometry Probability

Programming:

C++ C Python Lua

Research:

Benchmark Design Experiment Tracking
Technical Writing

/// EDUCATION

MSc Computer Graphics

Fictional Academy of Visual Computing

Status: Research completed

BSc Software Systems

Imaginary Polytechnic

Status: First Class

Advanced Diploma in Programming

Northstar Technical College

Status: Completed

/// RESEARCH

Focus:

GPU resource scheduling
Material systems Visibility
Frame graphs

Methods:

Controlled benchmarks GPU captures
Micro-tests

Output:

Technical reports Reproducible scenes
Performance tables

/// TALKS

Event: Fictional Graphics Meetup

Topic: Thinking in GPU Resource Lifetimes

Event: Imaginary Rendering Workshop

Topic: From Pass Declarations to a Working Frame

/// PROJECTS

Visibility Atlas (Research Engineer)

Synthetic benchmark investigating visibility-buffer layouts across scenes with highly different triangle and material distributions.

C++ Vulkan Compute Benchmarking

Lumenless (Rendering Researcher)

Fictional renderer prototype comparing tiled lighting and clustered-light approaches under controlled workloads.

C++ GLSL Vulkan Lighting

Shader Observatory (Tooling Engineer)

Offline shader inspection tool that groups reflection metadata and reports resource usage per pipeline.

C++ SPIR-V Reflection

Mesh Census (Pipeline Developer)

Asset-analysis utility for measuring topology, UV density, material count, and texture footprint in synthetic datasets.

C++ Assimp CLI